



MYTHIC

Compiler Optimization Report - YOLOv8s Pose

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1 Summary

For YOLOv8s Pose at 1088x1920x3 resolution:

Performance simulation estimates a Total Processing Time (for the compiled ONNX file) of

- 21.92 ms, of which
- 7.02 ms is Analog NPU processing time, which corresponds to
 - 71.1% ACE utilization, using 24 Analog Compute Engines; and of which
- 14.91 ms is Digital NPU processing time,
 - using only 8 Digital NPU cores, which is sufficient to achieve 30 fps.

The number of used Digital NPU cores (8) was selected to achieve at least 30 fps. If many more Digital NPU cores were used, the Digital NPU processing time could be substantially shortened. For example, 100 Digital NPU cores could reduce the Digital NPU processing time by ~12x, resulting in a total (Analog + Digital) processing time of ~8.2 ms, corresponding to ~120 fps.

There are no digital bottlenecks in the Analog NPU processing time.

Automatic optimization performs as well as the best manual optimization.

We may be able to improve performance in the future by finding a more optimal placement of weight blocks to ACEs (to increase the ACE utilization), but note that the current ACE utilization of 71.1% is already quite high.

We may be able to improve performance in the future by allowing weight block widths to exceed 256. The hardware design recently increased support to 512, and the compiler will add support for 512 in the future.

The estimated power is

- 0.4 Watts at the target frame rate (30 fps), and
- 0.6 Watts at the maximum frame rate (45 fps), if restricted to using only 8 Digital NPU cores.

Power estimates are approximate, and will be more precise in the future.

2 Reproducing the results

The results in this report were produced using the `“high_optimization”` compiler config for YOLOv8s Pose.

Note that some variance in results is expected (up to a few percent), due to non-deterministic, multi-threaded compiler optimizations.

The `mythic-compiler` command was

```
mythic-compiler \  
  --input-artifact $MYTHIC_SDK_ROOT/models/training/yolov8/yolov8s-  
pose_1920x1088_trained.tar.gz \  
  --compiler-config $MYTHIC_SDK_ROOT/mythic-model-zoo/configs/yolov8/compiler/  
yolov8s_pose_1088x1920x3_m2024_high_optimization.yaml \  
  --output-artifact /rscratch/tonbomythic3/<your_username>/  
yolov8s_pose_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

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The `mythic-ppa-estimators` command for the maximum frame rate (45 fps) was

```
mythic-ppa-estimators \  
  --estimate-performance \  
  --estimate-power \  
  /rscratch/tonbomythic3/<your_username>/  
yolov8s_pose_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

The `mythic-ppa-estimators` command for the target frame rate (30 fps) was

```
mythic-ppa-estimators \  
  --estimate-performance \  
  --estimate-power \  
  --power-inference-rate 30 \  
  /rscratch/tonbomythic3/<your_username>/  
yolov8s_pose_1088x1920x3_m2024_high_optimization_compiled.tar.gz
```

3 Performance estimates

The `mythic-ppa-estimators` tool outputs the following performance statistics:

```
Analog NPU Total Estimated Processing Time for Compiled ONNX File: 7.02 ms  
Analog NPU Estimated Frame Rate for Compiled ONNX File: 142.52 fps (frames per second)  
Critical Path ACE Latency: 7.02 ms  
Maximum SRAM Read/Write Time (over 6 ACE tiles): 5.38 ms  
Maximum SIMD Operation Time (over 6 ACE tiles): 0.00 ms  
Average SRAM Read/Write Time (averaged over 6 ACE tiles): 4.44 ms  
Average SIMD Operation Time (averaged over 6 ACE tiles): 0.00 ms  
Total Executed ACE Operations: 748,442  
Total Executed ACE MACs: 109,121,107,968  
Maximum Theoretical ACE Execution Time (With No Parallelization Using 1 ACE): 119.75 ms  
Minimum Theoretical ACE Execution Time (With Even Parallelization Across 24 ACEs): 4.99  
ms  
ACE Utilization (Minimum Theoretical Time/Estimated Processing Time): 71.11%  
Total SRAM Bytes Read Across Chip: 1,917,506,632 bytes  
Total SRAM Bytes Written Across Chip: 978,981,208 bytes  
Estimated Die Area to Achieve Estimated Processing Time: 126.55 mm^2  
-----  
Digital NPU Performance Metrics (from separate ONNX graph processing)  
Total Estimated MACs Targeting Digital NPU: 2,786,000,000  
Digital NPU MAC Utilization: 36.49%  
Digital NPU Cores: 8  
Digital NPU Frequency: 1,000 MHz  
Digital Estimated Frame Processing Time: 14.91 ms
```

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Digital Estimated Frame Rate: 67.08 fps (frames per second)

Combined Analog + Digital NPU Latency: 21.92 ms

Combined Analog + Digital NPU Total Estimated Frame Rate: 45.61 fps (frames per second)

NOTE: The information reported above may include the Analog NPU or the Analog NPU and a digital NPU as indicated. This estimated processing time and associated frame rate only covers the execution of the compiled ONNX model(s) running on one or both of those compute solutions in isolation. It does not include PCIe transfer overhead. Performance estimation **for** additional processing steps and a combined compute hardware solution will be added in future versions.

4 Power estimates

The `mythic-ppa-estimators` tool outputs the following power estimates.

Power at the target frame rate (30 fps):

Process Nodes: Analog 28nm, Digital 5nm

Number of ACEs: 24

Number of Digital NPU Cores: 8

Inference Rate: 30 frames/second (fps)

Analog NPU Estimated Power: 0.321 W **for** target ONNX file

Functional Unit Power: 0.262 W

Interconnect Power: 0.059 W

Digital NPU Estimated Power: 0.071 W **for** digital target ONNX file

Total Combined (Analog + Digital NPU) Power: 0.392 W

NOTE: Estimation **for** leakage, clock tree, chip I/O power will be added in future versions.

Power at the maximum frame rate (45 fps):

Process Nodes: Analog 28nm, Digital 5nm

Number of ACEs: 24

Number of Digital NPU Cores: 8

Inference Rate: 45 frames/second (fps)

Analog NPU Estimated Power: 0.450 W **for** target ONNX file

Functional Unit Power: 0.362 W

Interconnect Power: 0.089 W

Digital NPU Estimated Power: 0.106 W for digital target ONNX file

Total Combined (Analog + Digital NPU) Power: 0.556 W

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future versions.

5 Compiler reports

5.1 L0 IR stats

From near the end of mythic-compiler stdout:

```
===== L0 IR stats =====  
      Tiles: 7  
      Buffers: 162  
      Launchers: 245  
      ACE Launchers: 101  
      ACE Weights: 17023336  
      ACE calculations: 748442  
      Max ACE calculations: 130560 on tile 1_2  
=====
```

5.2 ACE utilization report

The compiler-generated `ace_utilization.txt`:

Summary:

- 81.63% ACE utilization is predicted by the compiler,
- assuming no digital bottlenecks (which can typically be eliminated).
- 748,442 ACE calculations are spread across 24 ACEs, which is
- 31,185.08 ACE calculations per ACE (on average).
- 38,204 ACE calculations are in the critical path, implying
- 6.11 ms predicted inference time (160 ns per ACE calculation).

Tile	ACEs	ACE calculations	ACE calculations Per ACE	Predicted Full Network ACE Critical Path	Predicted ACE Utilization
1_1	0	N/A	N/A	N/A	N/A
3_1	4	124,440	31,110.00	38,204	81.43%
4_1	4	129,472	32,368.00	38,204	84.72%
1_2	4	130,560	32,640.00	38,204	85.44%
2_2	4	128,622	32,155.50	38,204	84.17%
3_2	4	118,558	29,639.50	38,204	77.58%
4_2	4	116,790	29,197.50	38,204	76.43%
All	24	748,442	31,185.08	38,204	81.63%

5.3 Weight utilization report

The compiler-generated `weight_utilization.txt`:

Summary:

- 56.96% of analog weight storage capacity is used, using
- 17,023,336 weights out of 29,884,416 capacity, using
- 101 weight blocks.

Tile	Weight Blocks	Allocated Weights	Weight Capacity	Utilization
3_1	14	2,129,168	4,980,736	42.75%
4_1	17	2,827,192	4,980,736	56.76%
1_2	18	2,793,872	4,980,736	56.09%
2_2	15	2,419,840	4,980,736	48.58%
3_2	17	2,793,480	4,980,736	56.09%
4_2	20	4,059,784	4,980,736	81.51%
All	101	17,023,336	29,884,416	56.96%

5.4 SRAM utilization report

The compiler-generated `sram_utilization.txt`:

Summary:

- 4,775,962 bytes of SRAM is used, consisting of
 - 4,480,512 bytes **for** buffers (93.81%) and
 - 295,450 bytes **for** control flow (6.19%), using
 - 18.98% of the simulated on-chip SRAM capacity.

Tile	Buffer Allocation (bytes)	Control Flow Allocation (bytes)	Total Allocation (bytes)	SRAM Capacity (bytes)	Utilization
3_1	810,336	41,887	852,223	4,194,304	20.32%
4_1	91,328	21,681	113,009	4,194,304	2.69%
1_2	333,376	50,094	383,470	4,194,304	9.14%
2_2	1,909,280	48,324	1,957,604	4,194,304	46.67%
3_2	635,552	57,780	693,332	4,194,304	16.53%
4_2	700,640	75,684	776,324	4,194,304	18.51%
All	4,480,512	295,450	4,775,962	25,165,824	18.98%